

A REPRESENTATION THEORETIC PROOF OF BURNSIDE'S THEOREM

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Introduction

These notes were written as a bonus-question write-up for PMATH 348 (Fields and Galois Theory) at the University of Waterloo. The goal is to present a proof of Burnside's Theorem using the basic character theory of finite groups.

In these notes we develop the basic theory of characters of finite groups needed to prove **Burnside's Theorem**: *every finite group whose order has the form $p^a q^b$, where p and q are primes, is solvable.*

None of the results presented here are original. The exposition and overall structure follow very closely the notes of A. Zimmerman [1], with minor adjustments in notation and presentation.

Definition. Let G be a group. A representation of G is a pair (V, ρ) where V is a $\mathbb{C}$ -vector space and $\rho : G \rightarrow \text{GL}(V)$ is a homomorphism.

Theorem 1. *Let $\rho : G \rightarrow \text{GL}(V)$ is a representation of a finite group G . If V' is a subrepresentation of V , then there exists a complement W , of V' that is also a subrepresentation of V .*

Proof. Fix any hermitian inner product $\langle \cdot, \cdot \rangle : V \rightarrow \mathbb{C}$. Indeed such inner product exists since V is a finite dimensional $\mathbb{C}$ -vector space, for example the complex dot-product. Then the map $\langle \cdot, \cdot \rangle_0 : V \rightarrow \mathbb{C}$

$$\langle x, y \rangle_0 = \frac{1}{|G|} \sum_{g \in G} \langle \rho_g(x), \rho_g(y) \rangle$$

is also an Hermitian inner product. Let $W = (V')^\perp$ be the orthogonal complement of V' . We now show W is stable under the action of G , i.e, for all $h \in G$, we have that

$\rho_h(W) \subseteq W$. Let $v' \in V'$, $w \in W$ and $h \in G$. Then

$$\begin{aligned} \langle \rho_h(w), v' \rangle_0 &= \frac{1}{|G|} \sum_{g \in G} \langle \rho_g(\rho_h(w)), \rho_g(v') \rangle = \frac{1}{|G|} \sum_{g \in G} \langle \rho_{gh}(w), \rho_{ghh^{-1}}(v') \rangle \\ &= \frac{1}{|G|} \sum_{g \in G} \langle \rho_{gh}(w), \rho_{gh}(\rho_{h^{-1}}(v')) \rangle \end{aligned}$$

Indeed the right action of h on G is a bijection $G \rightarrow G$, hence as g runs over G , so does gh . Therefore the inner product becomes,

$$\langle \rho_h(w), v' \rangle_0 = \frac{1}{|G|} \sum_{g \in G} \langle \rho_g(w), \rho_g(\rho_{h^{-1}}(v')) \rangle = \langle w, \rho_{h^{-1}}(v') \rangle_0.$$

Since $(V', \rho|_{V'})$ is a subrepresentation, we have that $\rho_g(V') \subseteq V'$ for all $g \in G$ and so $\rho_g(\rho_{h^{-1}}(v')) \in V'$. Since W is the orthogonal complement, we must have that $\langle w, \rho_{h^{-1}}(v') \rangle_0 = 0$. Hence $\rho_h(w) \in (V')^\perp = W$, and so $(W, \rho|_W)$ is a subrepresentation of (V, ρ) . 

Corollary. Every representation of finite degree of a finite group is isomorphic to a direct sum of irreducible representations.

Proof. We proceed induction. Indeed if the representation has degree 1, then it is irreducible since any subspace of $\mathbb{C}$ is $\mathbb{C}$. Now suppose (V, ρ) is of degree n . If V is irreducible then we are done. Otherwise there exists a subrepresentation V' . By the previous theorem, $(V')^\perp$ is also a subrepresentation of V with respect to some inner product. By the induction hypothesis, both V' and $(V')^\perp$ are isomorphic to direct sums of irreducible subrepresentations. Hence $V \cong V' \oplus (V')^\perp$ is as well. 

Definition. The character of a representation $\rho : G \rightarrow \text{GL}(V)$ is the map $\chi : G \rightarrow \mathbb{C}$ defined as $\chi(g) = \text{Tr}(\rho_g)$.

Proposition 2. Let χ be a character of a representation of a finite group G of degree n . Then for all $g, h \in G$, we have that

1. $\chi(1) = n$
2. $\chi(g^{-1}) = \overline{\chi(g)}$
3. $\chi(gh) = \chi(hg)$.

Proof. (1) Indeed ρ_1 is the $n \times n$ identity matrix which has trace n . 

Proof. (2) Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of ρ_g . Then eigenvalues of $\rho_{g^{-1}}$ are the reciprocals of the λ_i 's. Moreover, since G is finite, the order of ρ_g is finite, and so the eigenvalues of ρ_g are roots of unity. Hence $1/\lambda_i = \overline{\lambda_i}$ and so the eigenvalues of $\rho_{g^{-1}}$ are

the complex conjugates of the eigenvalues of ρ_g . Therefore

$$\chi(g^{-1}) = \sum_{i=1}^n \bar{\lambda}_i = \overline{\sum_{i=1}^n \lambda_i} = \overline{\chi(g)}.$$

This prove (2). 

Proof. (3) The result follows from the fact that for any matrices A and B , the trace of their products satisfy, $\text{Tr}(AB) = \text{Tr}(BA)$. 

Definition. Let G be a group and $\rho^1 : G \rightarrow \text{GL}(V_1)$ and $\rho^2 : G \rightarrow \text{GL}(V_2)$ be representations of G . We define $V_1 \oplus V_2$ as a representation ρ by setting $\rho_g = \rho_g^1 \oplus \rho_g^2$ for all $g \in G$

Proposition 3. Let (V_1, ρ^1) and (V_2, ρ^2) be representations of a group G and χ_1 and χ_2 be their characters respectively. Then the character of $V_1 \oplus V_2$ is $\chi_1 + \chi_2$.

Proof. Let $g \in G$. Let $[\cdot]$ denote the matrix of a linear operator with respect to any basis. Then

$$[\rho_g] = \begin{pmatrix} [\rho_g^1] & 0 \\ 0 & [\rho_g^2] \end{pmatrix}$$

which has trace $\chi_1(g) + \chi_2(g)$ 

Definition. A homomorphism bewtween representations (V_1, ρ^1) and (V_2, ρ^2) is a vector space homomorphism $\varphi : V_1 \rightarrow V_2$ such that $\varphi \circ \rho_g^1 = \rho_g^2 \circ \varphi$, i.e, the following diagram commutes

$$\begin{array}{ccc} V_1 & \xrightarrow{\rho_g^1} & V_1 \\ \varphi \downarrow & & \downarrow \varphi \\ V_2 & \xrightarrow{\rho_g^2} & V_2 \end{array}$$

We say the two representations are isomorphic if and only if φ is an isomorphism, and denote $(V_1, \rho^1) \cong (V_2, \rho^2)$

Theorem 4. [Schur's Lemma] Let G be a group and (V_1, ρ_1) , and (V_2, ρ_2) be irreducible representations of G . If $\varphi : V_1 \rightarrow V_2$ is a homomorphism of representations, then

1. If $(V_1, \rho^1) \not\cong (V_2, \rho^2)$, then $\varphi = 0$
2. If $V_1 = V_2$, then $\varphi = \lambda I$ for some $\lambda \in \mathbb{C}$.

Proof. (1) Suppose $\varphi \neq 0$. We show φ is an isomorphism. We first show it is injective. Since it is not the zero map, we have $\ker \varphi \neq V_1$. If $w \in \ker \varphi$ then

$$\varphi \circ \rho_g^1(w) = \rho_g^2 \circ \varphi(w) = \rho_g^2(0) = 0.$$

Therefore $\rho_g^1(w) \in \ker \varphi$ and so $(\ker \varphi, \rho^1)$ is a subrepresentation of (V_1, ρ^1) . Since (V_1, ρ^1) is irreducible, and $\ker \varphi \neq V_1$, we must have that $\ker \varphi = \{0\}$. Hence φ is injective. We

now show surjectivity. Indeed since $\varphi \neq 0$, we have that $\text{im } \varphi \neq 0$. Similarly, if $w \in \text{im } \varphi$, then $w = \varphi(v)$ for some $v \in V_1$ and

$$\rho_g^2(w) = \rho_g^2 \circ \varphi(v) = \varphi \circ \rho_g^1(v) = \varphi(\rho_g^1(v)),$$

and so $\rho_g^2(w)$ is in the image of φ . Hence the image of φ is a subrepresentation of V_2 . Since the image is not zero, we must have that the image is all of V_2 and so φ is surjective. Hence it is an isomorphism of representations, and so $(V_1, \rho^1) \cong (V_2, \rho^2)$, a contradiction. 

Proof. (2) Let λ be an eigenvalue of φ and $\psi = \varphi - \lambda I$. An eigenvalue exists since V_1 is a $\mathbb{C}$ -vector space. One checks in private that ψ is a homomorphism of representations. Indeed the kernel is not trivial since if v is non-zero an eigenvector of φ , then $\psi(v) = 0$. Since V_1 is irreducible, ψ must be the zero map, and so $\varphi = \lambda I$. 

Definition. Let G be a finite group. For any functions $\phi, \psi : G \rightarrow \mathbb{C}$ we denote

$$\langle \phi, \psi \rangle := \frac{1}{|G|} \sum_{g \in G} \phi(g) \psi(g^{-1})$$

and call it the convolution of ϕ and ψ .

Lemma 5. Let (V, ρ) and (W, π) be representations of a finite group G and let $T : V \rightarrow W$ be a linear map. Define

$$T_0 := \frac{1}{|G|} \sum_{g \in G} \pi_g T \rho_{g^{-1}}.$$

Then T_0 is a homomorphism of representations, i.e., $T_0 \circ \rho_h = \pi_h \circ T_0$ for all $h \in G$.

Proof. Let $h \in G$. Then

$$\pi_h T_0 = \frac{1}{|G|} \sum_{g \in G} \pi_h \pi_g T \rho_{g^{-1}} = \frac{1}{|G|} \sum_{g \in G} \pi_{hg} T \rho_{g^{-1}}.$$

Indeed the left action of h on G is a bijection $G \rightarrow G$, hence as g runs over G , so does hg . Thus

$$\pi_h T_0 = \frac{1}{|G|} \sum_{g \in G} \pi_g T \rho_{g^{-1}h} = \left(\frac{1}{|G|} \sum_{g \in G} \pi_g T \rho_{g^{-1}} \right) \rho_h = T_0 \rho_h.$$

Hence $T_0 \circ \rho_h = \pi_h \circ T_0$ for all $h \in G$. 

Theorem 6. Let (V, ρ) and (W, π) be irreducible representations of a finite group G , and let χ_ρ and χ_π be their characters. Then

$$\langle \chi_\rho, \chi_\pi \rangle = \begin{cases} 1 & \text{if } (V, \rho) \cong (W, \pi), \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Let $T : V \rightarrow W$ be any linear map and define

$$T_0 := \frac{1}{|G|} \sum_{g \in G} \pi_g T \rho_{g^{-1}}.$$

By Theorem 5, T_0 is a homomorphism of representations, i.e. $T_0 \circ \rho_h = \pi_h \circ T_0$ for all $h \in G$. In particular $T_0 \in \text{Hom}_G(V, W)$. Moreover, if $T \in \text{Hom}_G(V, W)$ already, then for every $g \in G$ we have $\pi_g T \rho_{g^{-1}} = T$, and so $T_0 = T$. Hence the assignment $P : \text{Hom}_{\mathbb{C}}(V, W) \rightarrow \text{Hom}_{\mathbb{C}}(V, W)$, $P(T) = T_0$, is a projection with image exactly $\text{Hom}_G(V, W)$. Therefore

$$\text{Tr}(P) = \dim(\text{im } P) = \dim \text{Hom}_G(V, W).$$

Fix a basis $(v_1, \dots, v_n)$ of V with dual basis $(f_1, \dots, f_n)$, and fix a basis $(w_1, \dots, w_m)$ of W . For $1 \leq a \leq m$ and $1 \leq b \leq n$ define $E_{ab} \in \text{Hom}_{\mathbb{C}}(V, W)$ by

$$E_{ab}(v_b) = w_a, \quad E_{ab}(v_j) = 0 \ (j \neq b).$$

Then $\{E_{ab}\}_{a,b}$ is a basis of $\text{Hom}_{\mathbb{C}}(V, W)$. For $g \in G$ consider the operator $A_g : \text{Hom}_{\mathbb{C}}(V, W) \rightarrow \text{Hom}_{\mathbb{C}}(V, W)$ given by $A_g(T) = \pi_g T \rho_{g^{-1}}$, so that

$$P = \frac{1}{|G|} \sum_{g \in G} A_g.$$

Write $[\pi_g] = (p_{ij})$ and $[\rho_{g^{-1}}] = (r_{ij})$ in the chosen bases. A direct computation shows

$$A_g(E_{ab}) = \sum_{i=1}^m \sum_{j=1}^n p_{ia} r_{bj} E_{ij},$$

and so the coefficient of E_{ab} in $A_g(E_{ab})$ is $p_{aa} r_{bb}$. Hence

$$\text{Tr}(A_g) = \sum_{a=1}^m \sum_{b=1}^n p_{aa} r_{bb} = \left(\sum_{a=1}^m p_{aa} \right) \left(\sum_{b=1}^n r_{bb} \right) = \text{Tr}(\pi_g) \text{Tr}(\rho_{g^{-1}}) = \chi_{\pi}(g) \chi_{\rho}(g^{-1}).$$

By linearity of trace we obtain

$$\text{Tr}(P) = \frac{1}{|G|} \sum_{g \in G} \text{Tr}(A_g) = \frac{1}{|G|} \sum_{g \in G} \chi_{\pi}(g) \chi_{\rho}(g^{-1}) = \langle \chi_{\rho}, \chi_{\pi} \rangle.$$

Thus $\langle \chi_{\rho}, \chi_{\pi} \rangle = \text{Tr}(P) = \dim \text{Hom}_G(V, W)$. Finally, by Schur's Lemma, $\dim \text{Hom}_G(V, W) = 0$ if $(V, \rho) \not\cong (W, \pi)$, and $\dim \text{Hom}_G(V, W) = 1$ if $(V, \rho) \cong (W, \pi)$. Therefore

$$\langle \chi_{\rho}, \chi_{\pi} \rangle = \begin{cases} 1 & \text{if } (V, \rho) \cong (W, \pi), \\ 0 & \text{otherwise,} \end{cases}$$

as desired. 

Theorem 7. *Let V be a representation of a finite group G with character ϕ and $V \cong W_1 \oplus W_2 \oplus \cdots \oplus W_n$ be a decomposition of irreducibles. Then, if W is any irreducible representation of G with character χ , then*

$$\#\{W_i : W \cong W_i\} = \langle \phi, \chi \rangle.$$

Proof. By Proposition 7, the character ϕ decomposes into $\phi = \chi_1 + \chi_2 + \cdots + \chi_n$. Therefore

$$\langle \phi, \chi \rangle = \sum_{i=1}^n \langle \chi_i, \chi \rangle.$$

By the previous theorem, $\langle \chi_i, \chi \rangle = 1$ if $W_i \cong W$ and 0 otherwise. Hence $\langle \phi, \chi \rangle$ is the number of such W_i . 

We now establish a convention. If a representation V decomposes into a direct sum of irreducible representations, for any irreducible factor W , we call the number of its occurrences in the direct sum the multiplicity of W . Moreover, for any positive integer n , we write $nW = W \oplus \cdots \oplus W$ n times.

Theorem 8. *If χ is the character of a representation V of a finite group G , then $\langle \chi, \chi \rangle = 1$ if and only if V is irreducible*

Proof. Let $V \cong n_1 W_1 \oplus n_2 W_2 \oplus \cdots \oplus n_r W_r$ be the irreducible decomposition of V where χ_i is the character of W_i . Then $\chi = n_1 \chi_1 + n_2 \chi_2 + \cdots + n_r \chi_r$, and so

$$\langle \chi, \chi \rangle = n_1 \langle \chi, \chi_1 \rangle + n_2 \langle \chi, \chi_2 \rangle + \cdots + n_r \langle \chi, \chi_r \rangle.$$

If $\langle \chi, \chi \rangle = 1$, then $r = 1$ and $n_1 = 1$, and thus V is irreducible. Conversely, if V is irreducible, then $\langle \chi, \chi \rangle = 1$. 

Definition. Let $V = \mathbb{C}^{|G|}$ with basis $\{e_g\}_{g \in G}$, and $\rho : G \rightarrow \text{GL}(V)$ given by $\rho_h(e_g) = e_{hg}$ for all $h \in G$.

Definition. A class function on a group G is a function $f : G \rightarrow \mathbb{C}$ such that $f(hgh^{-1}) = f(g)$ for all $g, h \in G$.

Definition. Let G be a finite group. We denote the set of isomorphism classes of the irreducible representations of G by $\hat{G}$. I.e

$$\hat{G} = \{[(V, \rho)]_{\cong} : (V, \rho) \text{ irreducible}\}.$$

Theorem 9. *Let H be the space of all class functions of G , and $(\cdot | \cdot) : H \times H \rightarrow \mathbb{C}$ be a hermitian inner product. Then the characters of all elements of $\hat{G}$ form an orthonormal basis for H .*

Proof. Indeed each character is a class function since, $\chi(hg) = \chi(gh)$, and so $\chi(hgh^{-1}) = \chi(g)$. Moreover, by theorem 12, they are orthonormal. Now suppose $f \in H$ is orthogonal to every character of the elements of $\hat{G}$. We now show that $f = 0$. For any irreducible representation $\rho : G \rightarrow \text{GL}_n(V)$, let

$$\rho_f = \sum_{g \in G} f(g) \rho_{g^{-1}}.$$

Then for all $h \in G$,

$$\rho_h^{-1} \rho_f \rho_h = \sum_{g \in G} f(g) \rho_h^{-1} \rho_{g^{-1}} \rho_g = \sum_{g \in G} f(h^{-1}gh) \rho_{h^{-1}} \rho_{g^{-1}} \rho_h = \rho_f$$

since f is a class function and conjugation by h is a bijection of G . Thus by Schur's Lemma, $f = \lambda I$ for some $\lambda \in \mathbb{C}$. Hence

$$n\lambda = \text{Tr}(\rho_f) = \sum_{g \in G} f(g) \text{Tr}(g^{-1}) = \sum_{g \in G} f(g) \chi(g^{-1}) = \langle f, \chi \rangle = 0.$$

Thus, $\rho_f = \lambda = 0$. Since any representation can be decomposed into a direct sum of irreducible representations, it follows that $\rho_f = 0$ even if ρ is not irreducible. If we take ρ to be the regular representation, then

$$0 = \rho_f e_1 = \sum_{g \in G} f(g) \rho_{h^{-1}} e_1 = \sum_{g \in G} f(g) e_{g^{-1}}.$$

Thus $\{e_g\}_{g \in G}$ is linearly independent, we must have that $f(g) = 0$ for all $g \in G$. ✶

Corollary. The size of $\hat{G}$ is the number of conjugacy classes of G .

Proof. Observe that the dimension H is exactly the number of conjugacy classes of G . By the above theorem, this is the number of isomorphism classes of irreducible representations of G which are determined by their characters. ✶

Proposition 10. Let G be a finite group. Let $h \in G$ and $\mathcal{O}(h)$ be the conjugacy class of h . Let $j \in G - \mathcal{O}(h)$ and $\chi_1, \dots, \chi_k$ be the characters of the elements of $\hat{G}$. Then

$$\sum_{i=1}^k \chi_i(h) \overline{\chi_i(h)} = \frac{|G|}{|\mathcal{O}(h)|}$$

and

$$\sum_{i=1}^k \chi_i(j) \overline{\chi_i(h)} = 0.$$

Proof. Let $f : G \rightarrow G$ be defined by

$$f(x) = \begin{cases} 1 & \text{if } x = h \\ 0 & \text{otherwise} \end{cases}$$

By the previous theorem, for all $g \in G$,

$$f(g) = \sum_{i=1}^l \langle f, \chi_i \rangle \chi_i(g) = \sum_{i=1}^k \frac{|\mathcal{O}(h)|}{|G|} \overline{\chi_i(h)} \chi_i(g) = \frac{|\mathcal{O}(h)|}{|G|} \sum_{i=1}^k \chi_i(g) \overline{\chi_i(h)}.$$

Then for any $g \in \mathcal{O}(h)$ then we get the first result, and for any $g \notin \mathcal{O}(h)$ we get the second result. 

Theorem 11. *Let G be a finite group, $\rho : G \rightarrow \text{GL}(V)$ be a representation of G , and χ be the character of (V, ρ) . Then, for all $g \in G$, the complex number $\chi(g)$ is an algebraic integer.*

Proof. Let $g \in G$. Since G is finite, ρ_g has finite order, say n . Then any eigenvalue λ of ρ_g , is an n -th root of unity. Since $\chi(g)$ is the sum of the eigenvalues of ρ_g , we must have that $\chi(g) \in \mathbb{Z}[\zeta_n]$. Hence $\mathbb{Z}[\chi(g)] \subseteq \mathbb{Z}[\zeta_n]$. Since $\mathbb{Z}[\zeta_n]$ is a finitely generated $\mathbb{Z}$ -module, we must have that $\mathbb{Z}[\chi(g)]$ is finitely generated. Hence $\chi(g)$ is an algebraic integer. 

Proposition 12. *Let G be a finite group and suppose $f \in Z(\mathbb{C}[G])$ has the form*

$$f = \sum_{g \in G} \alpha_g g$$

where each α_g is integral over $\mathbb{Z}$. If $\rho : G \rightarrow \text{GL}_n(V)$ is an irreducible representation with character χ , then

$$\frac{1}{n} \sum_{g \in G} \alpha_g \chi(g)$$

is integral over $\mathbb{Z}$.

Proof. We first show f is integral over $\mathbb{Z}$. Let $h \in G$. Since $f \in Z(\mathbb{C}[G])$, we have $hf = fh$, hence

$$h^{-1}fh = f.$$

Writing $f = \sum_{g \in G} \alpha_g g$, we compute

$$h^{-1}fh = \sum_{g \in G} \alpha_g h^{-1}gh.$$

Comparing coefficients in $\mathbb{C}[G]$ gives $\alpha_{h^{-1}gh} = \alpha_g$ for all $g, h \in G$, so the coefficients are constant on conjugacy classes. Let $\mathcal{O}(g_1), \dots, \mathcal{O}(g_k)$ be the distinct conjugacy classes

and set

$$s_i := \sum_{x \in \mathcal{O}(g_i)} x.$$

Then

$$f = \sum_{i=1}^k \alpha_i s_i$$

for some algebraic integers α_i . Consider

$$M := \mathbb{Z}s_1 \oplus \cdots \oplus \mathbb{Z}s_k \subseteq \mathbb{C}[G].$$

Products of class sums are again $\mathbb{Z}$ -linear combinations of class sums, so M is a subring of $\mathbb{C}[G]$ and is finitely generated as a $\mathbb{Z}$ -module. Thus each s_i is integral over $\mathbb{Z}$, and hence $f \in M$ is integral over $\mathbb{Z}$.

Now apply the representation. Since f is central, for every $h \in G$,

$$\rho(h)^{-1} \rho(f) \rho(h) = \rho(h^{-1} f h) = \rho(f),$$

so $\rho(f)$ commutes with $\rho(G)$. By Theorem 4, we have $\rho(f) = \lambda I$ for some $\lambda \in \mathbb{C}$. Taking traces,

$$n\lambda = \text{Tr}(\rho(f)) = \text{Tr}\left(\sum_{g \in G} \alpha_g \rho(g)\right) = \sum_{g \in G} \alpha_g \text{Tr}(\rho(g)) = \sum_{g \in G} \alpha_g \chi(g),$$

so

$$\lambda = \frac{1}{n} \sum_{g \in G} \alpha_g \chi(g).$$

Finally, since f is integral over $\mathbb{Z}$, the ring $\mathbb{Z}[f]$ is finitely generated as a $\mathbb{Z}$ -module. Applying ρ shows $\mathbb{Z}[\rho(f)]$ is finitely generated as a $\mathbb{Z}$ -module. But $\rho(f) = \lambda I$, so $\mathbb{Z}[\rho(f)]$ contains $\mathbb{Z}[\lambda]$ (via $z \mapsto zI$). Hence $\mathbb{Z}[\lambda]$ is finitely generated as an abelian group, so λ is an algebraic integer. 

Proposition 13. *If $\rho : G \rightarrow \text{GL}_n(V)$ is an irreducible representation of a finite group G , then $n \mid |G|$.*

Proof. Let χ be the character of ρ . Consider

$$f := \sum_{g \in G} \chi(g^{-1}) g \in \mathbb{C}[G].$$

Since χ is a class function, $f \in Z(\mathbb{C}[G])$. Also by Theorem 11, the coefficients $\chi(g^{-1})$ are algebraic integers. Applying Theorem 12 with $\alpha_g = \chi(g^{-1})$, we obtain that

$$\lambda := \frac{1}{n} \sum_{g \in G} \chi(g^{-1}) \chi(g)$$

is an algebraic integer. But using Theorem 2(2),

$$\sum_{g \in G} \chi(g^{-1}) \chi(g) = \sum_{g \in G} \overline{\chi(g)} \chi(g) = |G| \langle \chi, \chi \rangle.$$

Since ρ is irreducible, Theorem 8 gives $\langle \chi, \chi \rangle = 1$, hence

$$\lambda = \frac{|G|}{n}.$$

So $\frac{|G|}{n}$ is a rational algebraic integer, therefore an integer, and $n \mid |G|$. ✶

Burnside's Theorem

Lemma 14. *If $|G| = p^a$ for some prime p , then G is solvable.*

Proof. We induct on a . If $a = 0$ then G is trivial. Assume $a \geq 1$. If $Z(G) \neq \{1\}$, then $Z(G)$ is abelian, hence solvable. Also $|G/Z(G)| = p^{a'}$ for some $a' < a$, so by induction $G/Z(G)$ is solvable. Since solvability passes from a normal subgroup and quotient to the whole group, G is solvable.

So assume $Z(G) = \{1\}$. The class equation gives

$$|G| = |Z(G)| + \sum_i |\mathcal{O}(g_i)| = 1 + \sum_i |\mathcal{O}(g_i)|.$$

Each $|\mathcal{O}(g_i)|$ divides $|G| = p^a$, hence each is a power of p , so the right side is 1 plus a multiple of p . This cannot equal p^a . Contradiction. Therefore $Z(G) \neq \{1\}$ and the previous paragraph applies. ✶

Lemma 15. *Let $h \in G$ be nonidentity and suppose $|\mathcal{O}(h)| = p^a$ for some prime p . Then there exists an irreducible representation $\rho : G \rightarrow \text{GL}(V)$ with kernel $N \neq G$ such that $\rho(h)$ lies in the center of $\text{im}(\rho)$.*

Proof. Let $\chi_1, \dots, \chi_k$ be the irreducible characters, with χ_1 the trivial character. Apply with the pair $(j, h) = (1, h)$ (note that $1 \notin \mathcal{O}(h)$ since $h \neq 1$):

$$\sum_{i=1}^k \chi_i(1) \overline{\chi_i(h)} = 0.$$

Since $\chi_1(1) = 1$ and $\chi_1(h) = 1$, this gives

$$1 + \sum_{i \geq 2} \chi_i(1) \overline{\chi_i(h)} = 0.$$

We claim there exists some $i \geq 2$ such that $\chi_i(h) \neq 0$ and $p \nmid \chi_i(1)$. Indeed, if for every $i \geq 2$ with $\chi_i(h) \neq 0$ we had $p \mid \chi_i(1)$, then rearranging would show that $-\frac{1}{p}$ is a $\mathbb{Z}$ -linear

combination of algebraic integers $\chi_i(h)$ (using Theorem 11), hence would be an algebraic integer. But $-\frac{1}{p}$ is not an algebraic integer. Contradiction.

Fix such an irreducible character χ , and let ρ be an irreducible representation with character χ . Since χ is not the trivial character, ρ is nontrivial and $N := \ker \rho \neq G$. Define the class function $v : G \rightarrow \mathbb{C}$ by

$$v(g) = \begin{cases} 1 & g \in \mathcal{O}(h), \\ 0 & \text{otherwise.} \end{cases}$$

Consider

$$f := \sum_{g \in G} v(g^{-1}) g \in Z(\mathbb{C}[G]).$$

Applying Theorem 12 to this f and the irreducible ρ shows that

$$\frac{1}{\chi(1)} \sum_{g \in G} v(g^{-1}) \chi(g) = \frac{|\mathcal{O}(h)|}{\chi(1)} \chi(h)$$

is an algebraic integer. Since $|\mathcal{O}(h)| = p^a$ and $p \nmid \chi(1)$, this forces $\chi(1) \mid \chi(h)$ in the sense that $\chi(h)/\chi(1)$ is again an algebraic integer. In particular $\chi(h) \neq 0$.

Now $\chi(h)$ is a sum of $\chi(1)$ roots of unity (the eigenvalues of $\rho(h)$), so

$$|\chi(h)| \leq \chi(1),$$

with equality only if all eigenvalues are equal. Since $\chi(1)$ divides $\chi(h)$ and $\chi(h) \neq 0$, we must have $|\chi(h)| = \chi(1)$, hence all eigenvalues of $\rho(h)$ coincide. Therefore $\rho(h) = \lambda I$ is a scalar matrix, so $\rho(h)$ lies in the center of $\text{im}(\rho)$. 

Theorem 16 (Burnside). *Every group of order $p^a q^b$, where p, q are primes, is solvable.*

Proof. We induct on $|G|$. If $a = 0$ or $b = 0$, then $|G|$ is a prime power and G is solvable by Theorem 14. Assume $a, b \geq 1$. If $Z(G) \neq \{1\}$, then $Z(G)$ is abelian, hence solvable. Also $|G/Z(G)| < |G|$ and still has the form $p^{a'} q^{b'}$, so by induction $G/Z(G)$ is solvable. Hence G is solvable.

Now assume $Z(G) = \{1\}$. Then there exists a nonidentity element $h \in G$ such that $q \nmid |\mathcal{O}(h)|$. Indeed, if q divided $|\mathcal{O}(x)|$ for every $x \neq 1$, then the class equation

$$|G| = 1 + \sum_{x \neq 1} |\mathcal{O}(x)|$$

would imply $|G| \equiv 1 \pmod{q}$, contradicting $q \mid |G|$. Since $|\mathcal{O}(h)|$ divides $|G| = p^a q^b$ and is not divisible by q , we have $|\mathcal{O}(h)| = p^{a'}$ for some a' . By Theorem 15, there exists an irreducible representation ρ with kernel $N \neq G$ such that $\rho(h)$ is central in $\text{im}(\rho)$. If $N = \{1\}$ then ρ is faithful, so $G \cong \text{im}(\rho)$ and $\rho(h)$ central would imply h is central in G , contradicting $Z(G) = \{1\}$. Therefore N is a nontrivial proper normal subgroup. Both N

and G/N have strictly smaller order than G and still have order of the form $p^{a''}q^{b''}$, so by induction both are solvable. Hence G is solvable. 

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